

Waste Segregation

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Abstract—Waste management is one of the major challenges faced by both developed and developing countries. Improper disposal of waste leads to environmental pollution, foul odour, and the spread of diseases, making proper segregation very important. With the increase in population and industrialisation, the need for efficient and automated waste management systems has become essential. This project presents an automated waste segregation system that uses sensors, machine learning, and robotic mechanisms to classify different types of waste. A machine learning model is used to detect sanitary pads with an accuracy of approximately 90% using image processing techniques. Metallic waste is separated using a robotic arm integrated with an electromagnet, while wet waste is detected using sensors immediately after the conveyor stage. The remaining waste is directed to a separate bin. A shaker mechanism is used to ensure that waste items move one at a time, improving detection accuracy and reducing errors. The system provides a low-cost (₹8,500 prototype), efficient, and scalable solution for improving waste segregation and supporting better recycling practices.

Keywords—Automated Waste Segregation, Conveyor Belt System, Robotic Arm, Flap Mechanism, Metal Sensor, Waste Management, Smart Waste System

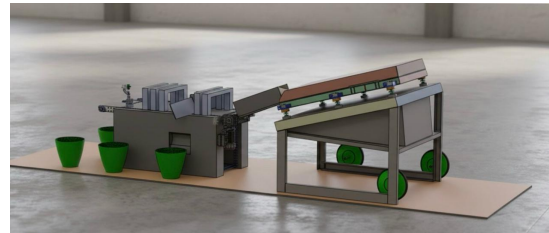
Introduction

Nowadays, waste management has become an important issue because of increasing population and urban development. Large amounts of waste are generated every day, and if it is not handled properly, it can cause pollution, bad smell, and health problems. Waste thrown in open areas also creates unhygienic conditions, so proper segregation and disposal are necessary. One major problem is that different types of waste are mixed together, such as wet waste, sanitary waste, and metal waste. This makes recycling difficult and reduces efficiency. Most traditional systems depend on manual sorting, which takes time and can be unsafe for workers.

In this project, an automated system is developed to improve waste segregation. The system uses a moisture sensor to detect wet waste just after the conveyor stage. A machine learning model is used to identify sanitary pads with around 90% accuracy using image

processing. Metal waste is separated using a robotic arm with an electromagnet, and the remaining waste is directed to another bin.

A shaker mechanism is also used to ensure that waste items move one by one, which helps in better detection. This system helps reduce manual work and makes waste sorting more efficient and practical.



I.

LITERATURE SURVEY

In recent years, waste management has become an important topic because the amount of waste generated in cities is increasing day by day. Many researchers have tried to improve waste segregation by using automation and smart technologies instead of depending only on manual work.

Some studies have shown that artificial intelligence and image processing can be used to identify different types of waste. In such systems, a camera captures images, and a machine learning model is used to classify the waste into categories. This method helps in improving accuracy and reducing human effort. However, these systems usually work only for limited types of waste and may not perform well in real-time conditions.

Other approaches focus on using sensors for waste detection. For example, moisture sensors are commonly used to identify wet waste, while other sensors detect different material properties. These systems are simple and low-cost, but they are not always able to detect specific types of waste like sanitary products.

There has also been work on using robotic arms for waste handling. These systems can pick and place waste based on detection results, which reduces manual effort and improves safety. In some cases, magnets are used to separate metal waste efficiently.

From all these studies, it can be understood that combining machine learning, sensors, and robotic systems can improve waste segregation. However, some common problems like mixed waste, overlapping items, and detection errors still exist. In this project, these issues are addressed by using a shaker mechanism to allow single-item flow, a machine learning model for sanitary pad detection, sensor-based wet waste identification, and a robotic arm with an electromagnet for metal separation, making the system more efficient and practical.

II. METHODOGY

In this project, an automated system is developed to sort different types of waste in a step-by-step manner. The process begins when waste is placed at the input section of the setup.

At the initial stage, a **shaker mechanism** is used to control the flow of waste. Instead of allowing multiple items to move together, the shaker ensures that waste moves one by one. This is important because overlapping of waste can lead to incorrect detection in later stages. By maintaining proper spacing between items, the overall accuracy of the system improves.

After this, the waste is carried forward using a **conveyor belt system**. The conveyor provides a continuous and controlled movement of waste through different processing stages. It also ensures that each component of the system gets enough time to analyze the waste properly.

The first detection stage focuses on identifying **wet waste**. A **moisture sensor** is used to measure the moisture content of each waste item. The sensor outputs a value, and if it crosses a certain threshold, the waste is considered wet. Once detected, a sorting mechanism directs the waste into the wet waste bin at an early stage itself. This helps in reducing unnecessary processing in the later stages.

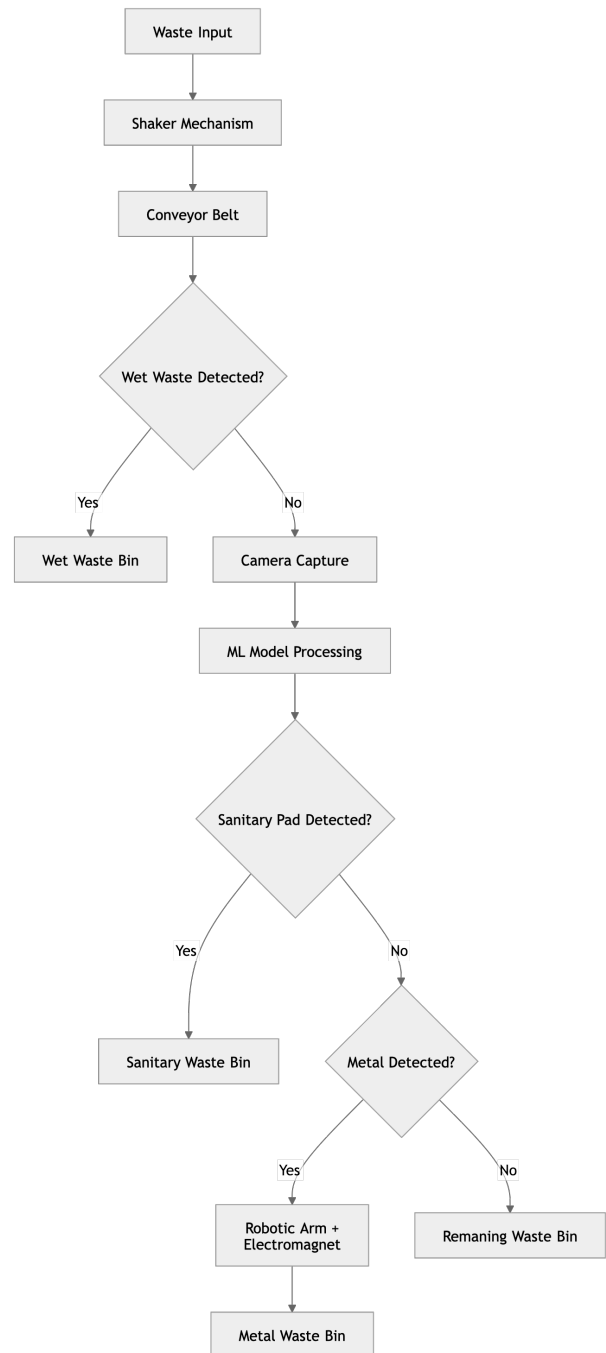
If the waste is not classified as wet, it moves to the next stage where a **camera module captures an image** of the object. This image is processed using a **machine learning model**, which has been trained to recognize sanitary pads. The model analyzes features from the image and gives a prediction. If the waste is identified as sanitary, it is separated into a dedicated sanitary waste bin. The model used in this system achieves an accuracy of around **90%**, which is suitable for practical applications.

If the waste is neither wet nor sanitary, the system proceeds to check for **metal content**. For this, a **robotic arm fitted with an electromagnet** is used. When a metal object is detected, the electromagnet is activated, and the robotic arm picks up the object. The arm then moves and places the metal waste into a separate metal bin. This mechanism helps in efficient and safe handling of metallic materials.

Any waste that does not fall into these categories is finally directed to a **general waste bin**. This ensures that all waste is sorted into appropriate categories based on its properties.

Overall, the system follows a clear sequence of operations, where each stage performs a specific task. The use of the shaker

mechanism, conveyor belt, sensors, machine learning model, and robotic arm together improves the efficiency of waste segregation. It also reduces human effort and makes the process more reliable and practical for real-world use.



III. RESULTS AND DISCUSSIONS

The proposed system was tested to check how well each part works in real conditions. The results show that the overall system is able to segregate different types of waste with good accuracy and consistency.

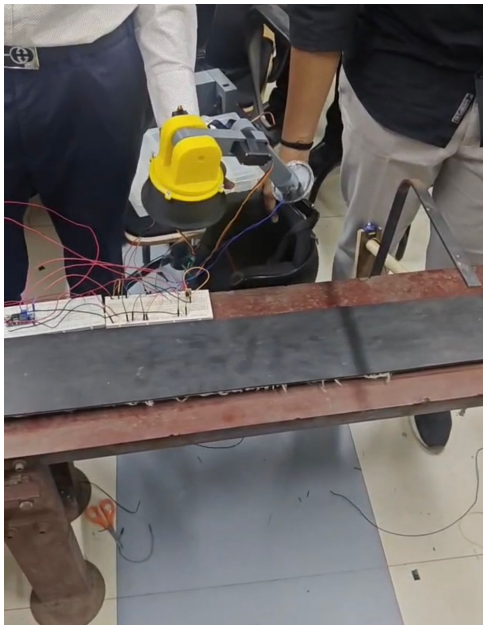
The machine learning model used for sanitary pad detection performed well during testing. It was able to identify sanitary waste with an accuracy of around 90%, which shows that image-based detection is effective for this type of application. Some minor errors were observed when the object was not clearly visible, but overall performance was satisfactory.

The wet waste detection stage also worked properly. The moisture sensor was able to correctly identify wet waste and separate it at an early stage. This helped reduce the load on later stages of the system and improved overall efficiency.

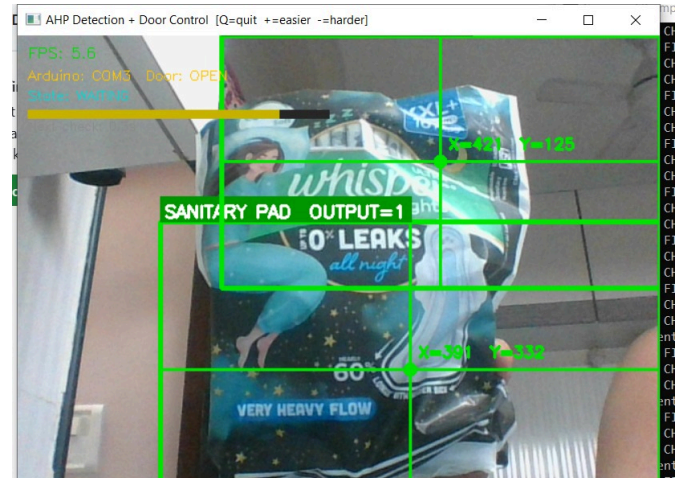
The robotic arm with electromagnet successfully detected and separated metallic waste. It was able to pick and place metal objects into the correct bin without major issues. The mechanism worked smoothly, although slight delays were noticed in arm movement during repeated testing.

The shaker mechanism played an important role in improving system performance. By allowing only one waste item at a time, it reduced overlapping and helped both sensors and the machine learning model give better results.

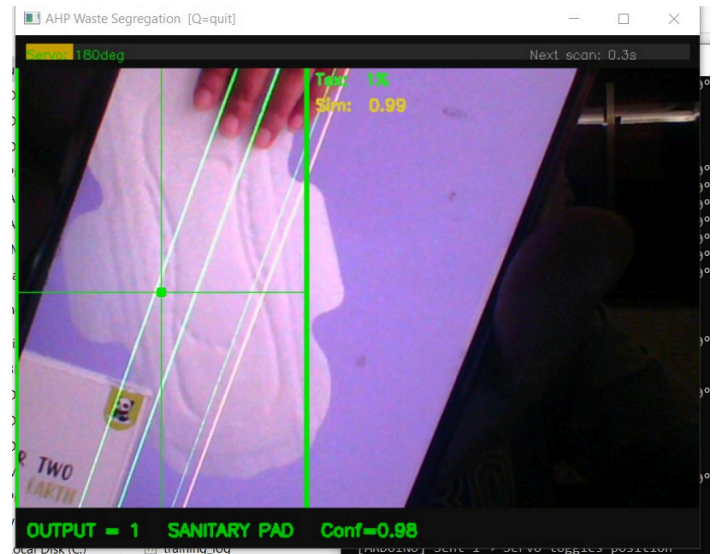
Overall, the system showed reliable performance and was able to reduce manual effort in waste segregation. The combination of sensors, machine learning, and mechanical components makes the system practical for real-world applications, especially where low-cost automated solutions are required.



A) Robotic Arm picking up and dropping the magnet in bin



B) WRAPPED SANITARY PAD DETECTION



C) SANITARY PAD DETECTION

CONCLUSION AND FUTURE SCOPE

In this project, an automated waste segregation system was designed and tested using a combination of sensors, machine learning, and mechanical components. The system was able to sort different types of waste such as wet waste, sanitary waste, and metal waste in a step-by-step manner. The results obtained during testing show that the system performs reliably under normal conditions.

The machine learning model used for sanitary pad detection achieved an accuracy of around 90%, which indicates that image-based classification can be effectively used for identifying specific types of waste. The moisture sensor was able to detect wet waste at an early stage, which helped reduce unnecessary processing in later stages. The robotic arm with an electromagnet successfully picked and separated metallic objects, showing that mechanical automation can be used for safe and efficient handling of waste. In addition, the shaker mechanism improved the overall system performance by ensuring that waste items moved one at a time, reducing errors caused by overlapping.

Another important outcome of this project is that the system is low-cost (approximately ₹8,500) and can be built using easily available components. This makes it suitable for small-scale applications such as homes, residential societies, schools, and public places where basic waste segregation is required. The system also helps in reducing manual effort and supports cleaner waste management practices.

For future improvements, the system can be enhanced in several ways. The machine learning model can be further trained to recognise more types of waste such as plastic bottles, paper, glass, and other recyclable materials. This will make the system more useful for complete waste segregation. The robotic arm can be upgraded with better control systems, faster movement, and improved gripping mechanisms to handle different shapes and sizes of waste.

More advanced sensors can be added to improve detection accuracy and reliability. The system can also be integrated with IoT (Internet of Things) so that data related to waste collection and bin status can be monitored remotely. This will help in better planning and management of waste collection.

In the long term, the system can be scaled and implemented in smart cities, industrial areas, and large waste management plants. With further development, such systems can play an important role in improving recycling efficiency, reducing environmental pollution, and creating cleaner and more sustainable surroundings.

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